



Screening Test and Its Validity

Lecture 11

Screening Test

- A screening test is done to detect potential health disorders or diseases in people who do not have any symptoms of disease.
- The goal is early detection and lifestyle changes , to reduce the risk of disease, or to detect it early enough to treat it most effectively.
- Screening refers to the application of a medical procedure or test to people who as yet have no symptoms of a particular disease, for the purpose of determining their likelihood of having the disease. The screening procedure itself does not diagnose the illness. Those who have a positive result from the screening test will need further evaluation with subsequent diagnostic tests or procedures.

Aims And Objectives

- To sort out from a large group of apparently healthy people, those likely to have disease or are at an increased risk.
- To bring those who are apparently abnormal under supervision, confirmation and treatment if required.

Validity Of Screening Tests

- The validity of a screening test is defined as its ability to distinguish between who has a disease and who does not and has 2 components:
 - Sensitivity : defined as ability of a test to identify correctly those who have the disease or true positives.
 - Specificity : defined as the ability of a test to identify correctly those who do not have a disease or true negatives.

Three Key Measures Of Validity

1. SENSITIVITY
2. SPECIFICITY
3. PREDICTIVE VALUE



Screening test results	Diagnosis		Total
	Present	Absent	
Positive	a (True positive)	b (False positive)	$a + b$
Negative	c (False negative)	d (True negative)	$c + d$
Total	$a + c$	$b + d$	$a + b + c + d$

Sensitivity

It is defined as the ability of a test to identify correctly all those who have the disease, that is "true positive".

$$\frac{(TP)}{(TP+FN)} \times 100 \text{ OR } \frac{(a)}{(a + c)} \times 100$$

- **A test has 90% sensitivity means:** 90 per cent of the diseased people screened by the test will give a "true positive" result and the remaining 10 per cent a "false negative" result.

Specificity

It is defined as the ability of a test to identify correctly those who do not have the disease, that is, "true negatives".

$$\frac{(TN)}{(FP+TN)} \times 100 \quad \text{OR} \quad \frac{(d)}{(b+d)} \times 100$$

- **90 per cent specificity means:** 90 per cent of the non diseased persons will give "true negative" result and 10 per cent of non-diseased people screened by the test will be wrongly classified as "diseased" when they are not (False positives).

Percentage Of False Positives:

$$\frac{(b)}{(b + d)} \times 100$$

- Means that patients who do not have the disease are told that they have the disease.
- Further diagnostic test.
- Inconvenience, discomfort, anxiety & unnecessary expense.

Percentage Of False Negatives:

$$\frac{(c)}{(a + c)} \times 100$$

- Means that patients who actually have the disease are told that they do not have the disease.
- False Re-assurance.
- Might ignore the development of signs & symptoms.
- Postpone treatment.
- **IMPORTANT IN SERIOUS DISEASE.**

Predictive Values Of A Test :

- Useful to know what proportion of patients with abnormal test results are truly abnormal.
- **They reflect the diagnostic power of a test.**

Positive Predictive Value:

The "predictive value of a positive test" indicates the probability that a patient with a positive test result has, in fact, the disease in question.

- Proportion of patients with positive test results who are correctly diagnosed.
- PPV of 90% means that 90% of the patients who are diagnosed to be positive by the test in fact have the disease in question.

$$\frac{(a)}{(a + b)} \times 100$$

Negative Predictive Value:

The "predictive value of a negative test" indicates the probability that a patient with a negative test result doesn't have disease in question.

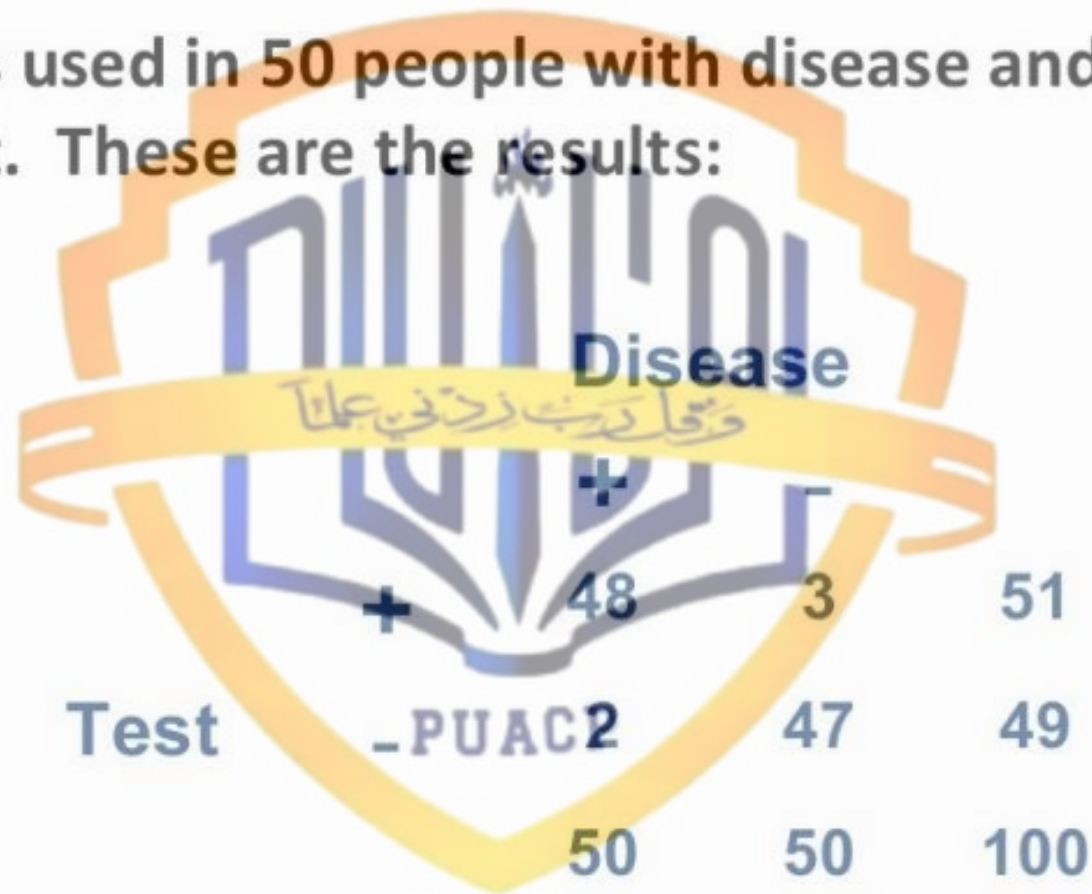
- NPV of 90% means that 90% of the patients who are diagnosed to be negative by the test do not have the disease in question.

$$\frac{(d)}{(c + d)} \times 100$$

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Calculating The Rates

A test is used in 50 people with disease and 50 people without. These are the results:



		Disease		
		+	-	
Test	+	48	3	51
	-	2	47	49
		50	50	100

		Disease		
		+	-	
Test	+	48	3	51
	-	2	47	49
		50	50	100

Sensitivity = $48/50 = 96\%$

Specificity = $47/50 = 94\%$

Positive predictive value = $48/51 = 94\%$

Negative predictive value = $47/49 = 96\%$

Percentage of False positive = $3/50 = 6\%$

Percentage of False negative = $2/50 = 4\%$

Effect of Prevalence :

- Predictive values depend strongly on prevalence of the condition.
- As the prevalence of the condition increases positive predictive value increases and thus more chances of getting true positive results.
- If the condition is uncommon it is more sure that the negative test indicates no abnormality.

Relationship B/W Predictive Value And Disease Prevalence

Relationship of Disease Prevalence to Positive Predictive Value					
EXAMPLE: SENSITIVITY = 99%, SPECIFICITY = 95%					
Disease Prevalence	Test Results	Sick	Not Sick	Totals	Positive Predictive Value
1%	+	99	495	594	$\frac{99}{594} = 17\%$
	-	1	9,405	9,406	
	Totals	100	9,900	10,000	
5%	+	495	475	970	$\frac{495}{970} = 51\%$
	-	5	9,025	9,030	
	Totals	500	9,500	10,000	

- Higher the prevalence, higher is the predictive value.
- Hence a screening program is most productive and efficient when it is directed to high risk target population.

Practice Question

- In a BCP Screening Test of 1600 cancer patients, the results are

Test\ Disease	Positive	Negative
Positive	570	150
Negative	30	850

- Compute the validity of screening test and discuss the results, where Test= BCP and disease = Cancer.